Practical no. 17

Aim - TestNG Framework Implementation and Reporting

Create and execute automated test cases using TestNG annotations. Group test cases, define priorities, and generate HTML test reports.

Step 1: Check Eclipse Java Version

In Eclipse: Window → Preferences → Java → Installed JREs

Select your JDK 21.0.12

Then go to: Java → Compiler

Set Compiler compliance level: 21

Click Apply and Close.

Step 2: Create Maven Project

File → New → Maven Project

Tick: Create a simple project (skip archetype selection) → Next

Group Id: TestNGSeleniumProject

Artifact Id: TestNGSeleniumProject

Packaging: jar

Click Finish.

Step 3: Install TestNG in Eclipse

Help → Eclipse Marketplace → Search: TestNG

Find: TestNG for Eclipse → Click Install

Follow the installation instructions and restart Eclipse when asked.

Check: Window → Preferences → TestNG (if TestNG appears, installation is successful).

Step 4: Open pom.xml

Replace its contents with:

```
<project>
    <modelVersion>4.0.0</modelVersion>
    <groupId>TestNGSeleniumProject</groupId>
    <artifactId>TestNGSeleniumProject</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <properties>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>4.35.0</version>
        </dependency>
        <dependency>
            <groupId>org.testng</groupId>
            <artifactId>testng</artifactId>
            <version>7.11.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

Save – Ctrl + S

Then: Right-click project → Maven → Update Project → OK

Wait until Maven finishes downloading the dependencies.

If the red error is still showing: Maven → Update Project..., then Project → Clean... → select TestNGSeleniumProject → Clean.

Step 5: Create TestNG Java Class

Right-click src/test/java → New → Package → Enter: testngtests

Right-click testngtests → New → Class → Enter: TestNGDemo → Finish

Step 6: Write the TestNG Program (TestNGDemo.java)

```
package testngtests;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

public class TestNGDemo {

    WebDriver driver;

    @BeforeMethod
    public void setup() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.get("https://www.google.com");
        System.out.println("Browser opened successfully");
    }

    @Test
    public void verifyTitle() {
        String title = driver.getTitle();
        System.out.println("Page Title: " + title);

        if (title.contains("Google")) {
            System.out.println("Test Passed");
        } else {
            System.out.println("Test Failed");
        }
    }

    @AfterMethod
    public void closeBrowser() {
        driver.quit();
        System.out.println("Browser closed");
    }
}
```

Step 7: Execute the Test

Right-click TestNGDemo.java → Run As → TestNG Test

If getting invalid version error: Run → Run Configurations → JRE tab → Execution environment: JavaSE-20 (jdk-21.0.12).

Step 8: Create Multiple Test Cases

Replace the Java code with:

```
package testngtests;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

public class TestNGDemo {
```

```
    WebDriver driver;

    @BeforeMethod
    public void setup() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.get("https://www.google.com");
        System.out.println("Browser opened");
    }

    @Test(priority = 1)
    public void verifyTitle() {
        String title = driver.getTitle();
        System.out.println("Title: " + title);

        if (title.contains("Google")) {
            System.out.println("Title Test Passed");
        } else {
            System.out.println("Title Test Failed");
        }
    }

    @Test(priority = 2)
    public void verifyURL() {
        String url = driver.getCurrentUrl();
        System.out.println("URL: " + url);

        if (url.contains("google")) {
            System.out.println("URL Test Passed");
        } else {
            System.out.println("URL Test Failed");
        }
    }

    @Test(priority = 3)
    public void verifyPageSource() {
        String source = driver.getPageSource();

        if (source.contains("Google")) {
            System.out.println("Page Source Test Passed");
        } else {
            System.out.println("Page Source Test Failed");
        }
    }

    @AfterMethod
    public void closeBrowser() {
        driver.quit();
        System.out.println("Browser closed");
    }
}
```

Step 9: Create TestNG XML File

Right-click the project → New → File → Name: testng.xml

Put this code inside:

```
<?xml version="1.0" encoding="UTF-8"?>
<suite name="TestNG Suite">
    <test name="Google Tests">
        <classes>
            <class name="testngtests.TestNGDemo"/>
        </classes>
    </test>
</suite>
```

Step 10: Run testng.xml

Right-click testng.xml → Run As → TestNG Suite

Step 11: Generate and Open HTML Report

Go to your project folder → test-output → index.html

Right-click index.html → open with web browser.

You can also open emailable-report.html for a simpler report.